

Alexander Britton

Applied Engineer

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Summary

Applied engineer building production systems across a wide variety of domains. 1.5 years developing computer vision pipelines, ML models, and data infrastructure. Published researcher with entrepreneurial background.

Experience

Driveline Baseball Tampa, FL
Forward Deployed Engineer Aug 2025 - Present

- Trained **YOLOv8 ball detection model** achieving mAP50 0.813 on 3,000+ annotated images with CVAT pipeline
- Built **3D pose estimation system** using multi-camera triangulation for markerless motion capture
- Architect **ML pipelines** processing terabytes of motion capture data across 3 production facilities
- Developed **MCP server** enabling natural language database queries via LLM integration
- Mentor technical staff and drive architecture decisions at highest-output facility

Driveline Baseball Tampa, FL
Sports Science Engineer June 2024 - Aug 2025

- Designed **two-stage ML pipeline** (XGBoost + neural net) for elbow torque prediction achieving $R^2=0.76$
- Embedded 34,000+ mocap trials into production MariaDB; designed ETL workflows for biomechanics data
- Built **motion capture frontend** controlling 8-camera system with session management
- Developed **CV calibration system** with extrinsics optimization and ground plane calibration

CerebriTech Orlando, FL
Founder 2021 - 2023

- Co-founded health-tech startup for real-time concussion detection using NASA-patented sensors
- **Top 5 Entrepreneur Under 25**, Orlando Business Journal (2022)

Projects

EMG Processing Pipeline
End-to-end pipeline with ML event detection, signal processing, and database integration
[GitHub](#)

RAG Document System
Production RAG system using HuggingFace transformers and vector DB with REST API
[GitHub](#)

Baseball Pitch Detection
YOLOv8 object detection model (mAP50 0.813) with CVAT annotation pipeline and trajectory scoring algorithm
[GitHub](#)

Education

Embry-Riddle Aeronautical University Bachelor of Science • 3.92 GPA, Summa Cum Laude
Mechanical Engineering Daytona Beach, FL • Aug 2020 - May 2024
Academic All-American | Division II Baseball (4-year starter)

Publications

Kinematic Analysis of a Four-Bar Linkage for Forward Stroke Motion March 2025
American Journal of Biomedical Science and Research
[View](#)

Neuromuscular Efficiency and Elbow Varus Torque in Baseball Pitchers 2025
Under Review

Profiles

[GitHub](#)
[GitHub](#)

Technical Skills

Languages
Python, JavaScript/TypeScript, SQL, MATLAB
● ● ● ● ●

Data
MariaDB, ETL pipelines, pandas, motion capture processing
● ● ● ● ●

ML / CV
YOLOv8, XGBoost, scikit-learn, PyTorch, CVAT, multi-camera triangulation
● ● ● ● ●

Infrastructure
Docker, Linux, server administration, LLM integration (LangChain)
● ● ● ● ○